

PANJ, Tajikistan

WMO: 389470

Lat: 37.233N

Lon: 69.083E

Elev: 1073

StdP: 14.14

Time Zone: 5.00 (E05)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.7	24.2	13.0	11.1	17.6	20.6	16.1	26.2	19.3	42.4	11.4	39.6	3.2	170	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.6	102.4	75.8	100.5	74.0	98.4	72.8	78.1	98.4	76.2	96.8	74.6	94.9	3.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	121.9	94.1	69.3	112.0	91.6	67.4	104.6	89.5	42.7	98.4	40.6	97.1	39.0	95.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.7	8.2	6.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	40.7	43.1	53.6	64.8	75.2	83.1	85.8	81.8	74.9	62.9	50.5	42.5	(6)
		DBStd	17.32	6.65	9.22	7.26	7.28	6.42	3.20	2.88	3.30	3.56	6.70	7.71	7.33	(7)
		HDD50	882	290	216	48	2	1	0	0	0	4	77	244		(8)
		HDD65	3077	753	612	357	90	13	0	0	2	117	435	698		(9)
		CDD50	5762	3	24	159	447	782	994	1110	987	748	404	93	11	(10)
		CDD65	2482	0	0	3	85	329	544	645	522	300	53	1	0	(11)
		CDH74	31858	0	3	61	815	3591	7259	9178	6765	3478	682	26	0	(12)
		CDH80	17801	0	0	8	253	1719	4268	5665	3961	1719	205	3	0	(13)
(14)	Wind	WSAvg	2.9	2.8	2.8	3.3	3.1	3.3	3.2	2.7	2.6	2.6	2.7	2.6	2.7	(14)
(15)	Precipitation	PrecAvg	19.0	2.2	3.3	3.3	2.7	1.8	0.6	0.6	0.5	0.3	0.6	1.4	1.6	(15)
(16)		PrecMax	25.0	5.6	6.7	5.8	4.1	5.1	1.9	1.7	1.0	1.2	1.7	4.2	3.4	(16)
(17)		PrecMin	10.6	0.8	1.3	0.6	0.9	0.2	0.1	0.1	0.1	0.0	0.3	0.3		(17)
(18)		PrecStd	3.5	1.1	1.3	1.3	0.9	1.0	0.4	0.4	0.2	0.3	0.5	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	61.4	72.3	80.5	90.7	98.8	103.6	105.5	102.4	96.4	89.7	77.6	65.7		(19)
(20)		0.4% MCWB	50.8	56.8	63.6	67.7	70.3	74.4	78.0	76.2	70.8	67.5	57.7	51.7		(20)
(21)		2% DB	57.4	65.8	74.3	86.1	95.4	101.0	103.3	99.8	94.4	85.3	71.6	61.0		(21)
(22)		2% MCWB	48.5	53.7	60.3	67.3	69.8	72.7	76.6	74.5	69.6	65.0	56.0	50.5		(22)
(23)		5% DB	54.3	60.7	70.1	82.5	92.7	98.8	101.6	98.0	91.7	81.4	67.3	57.1		(23)
(24)		5% MCWB	47.1	51.1	58.8	66.5	68.7	71.1	75.7	73.0	68.4	62.6	54.0	48.9		(24)
(25)		10% DB	51.3	56.6	66.1	78.5	89.5	96.6	99.6	95.7	89.3	77.1	63.3	53.8		(25)
(26)		10% MCWB	45.4	49.1	56.3	64.5	68.2	70.5	74.4	71.4	66.9	60.6	52.3	47.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	52.0	58.8	64.8	71.0	74.3	78.3	81.5	79.0	73.5	68.3	60.2	54.4		(27)
(28)		0.4% MCDB	59.6	70.9	78.3	85.3	92.8	98.1	101.0	97.9	93.2	86.2	71.9	61.5		(28)
(29)		2% WB	49.4	54.4	61.7	69.0	72.1	75.4	79.2	76.2	71.1	65.8	56.7	51.7		(29)
(30)		2% MCDB	56.1	63.7	71.7	82.7	90.3	96.5	99.5	96.3	90.9	83.1	69.5	57.3		(30)
(31)		5% WB	47.5	51.6	59.5	67.2	70.6	73.4	77.5	74.6	69.5	63.4	54.9	49.7		(31)
(32)		5% MCDB	53.4	59.5	68.6	79.9	88.4	95.3	98.3	94.6	89.2	79.1	65.8	55.4		(32)
(33)		10% WB	45.6	49.1	57.2	65.5	69.2	71.8	75.8	73.1	67.9	61.4	52.9	48.1		(33)
(34)		10% MCDB	50.5	56.1	65.3	77.3	86.4	92.9	96.5	92.5	87.0	75.6	62.8	53.4		(34)
(35)	Mean Daily Temperature Range	MDBR	15.4	17.6	18.5	20.6	24.0	27.2	27.6	28.4	29.5	25.0	20.2	15.6		(35)
(36)		5% DB MCDBR	22.3	25.6	25.7	26.8	29.6	30.7	30.7	30.2	32.9	30.7	27.3	23.5		(36)
(37)		5% DB MCWBR	15.1	16.0	14.2	13.2	12.5	12.9	14.7	13.6	15.8	15.3	14.3	14.9		(37)
(38)		5% WB MCDBR	20.2	23.4	23.6	24.9	26.9	29.3	29.4	29.0	32.2	28.9	24.9	19.9		(38)
(39)		5% WB MCWBR	14.3	15.4	13.9	13.2	13.2	13.9	15.1	14.1	16.2	15.2	14.0	13.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.403	0.460	0.514	0.525	0.538	0.571	0.555	0.519	0.484	0.513	0.476	0.419		(40)
(41)		taud	2.060	1.927	1.832	1.845	1.838	1.757	1.785	1.857	1.901	1.805	1.883	2.005		(41)
(42)		Ebn at Noon	235	236	236	243	242	234	237	242	242	215	205	217		(42)
(43)		Edh at Noon	39	51	62	64	66	71	69	63	57	57	47	39		(43)
(44)	All-Sky Solar Radiation	RadAvg	697	923	1355	1774	2203	2521	2516	2312	1945	1382	856	633		(44)
(45)		RadStd	73	115	111	158	141	98	75	60	44	96	94	82		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.65	-1.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page